

P1 - US Commercial Cage-Free Egg Production Compliance Context

Firm quantified standards, regulatory tripwires, and advisory guidance flags

Compiled from primary UEP Certified, FDA, AVMA, and USDA APHIS documents. Generated June 24, 2026.

UEP Certified cage-free standards are private certification standards with annual audits and explicit audit-failure triggers. FDA Egg Safety Rule requirements in 21 CFR Part 118 are binding federal regulation for covered shell-egg producers. AVMA depopulation guidance is professional guidance rather than law, but is operationally important because APHIS policies reference it during HPAI response. APHIS HPAI response and indemnity materials are federal outbreak-response policy and payment schedules.

The hard-stop items most relevant to an animal-welfare evaluation are: UEP space allowance, backfilling, commingling certified/non-certified eggs, feed-withdrawal molt, and willful abuse/neglect; FDA SE-positive environmental or egg-test diversion/treatment obligations; and APHIS approval/operational requirements when constrained-circumstances depopulation methods are used.

UEP Certified cage-free standards

Item	Firm value from primary standard	Binding status in UEP context	Range / contested flag	Source
Audit pass threshold	90% on annual UEP Certified audit	Firm certification threshold	No range	UEP-2024
Audit total points	200 total points; 180 required to pass in the public 2017 cage-free guideline	Historical public UEP audit metric; aligns with 90%	Current 2024 public guideline states 90% but does not publicly restate the 200-point score-sheet structure	UEP-2017; UEP-2024
Absolute audit-fail items	Space allowance; backfilling; commingling certified/non-certified eggs; feed-withdrawal molt; plus willful abuse/neglect in 2024 public guideline	Firm auto-fail triggers	No numeric range	UEP-2024; UEP-2017
Indoor floor space, multi-tier housing	1.0 sq ft/hen = 144 sq in/hen	Firm minimum	Conditional by system type	UEP-2024
Indoor floor space, slatted-floor housing	1.0 sq ft/hen = 144 sq in/hen	Firm minimum	Conditional by system type	UEP-2024
Indoor floor space, single-level all-litter housing	1.5 sq ft/hen = 216 sq in/hen	Firm minimum	Conditional by system type	UEP-2024
Litter share of total space	15% of total space for litter	Firm in 2017 public guideline	UEP noted more research may change this in future	UEP-2017
Ammonia target	<10 ppm	Advisory wording inside certification guideline ("should")	No numeric range, paired with ceiling below	UEP-2024
Ammonia ceiling	Must rarely exceed 25 ppm	Semi-hard upper-limit language in guideline	Rare-exceedance wording is not a simple continuous binary limit	UEP-2024
Layer-barn light minimum	0.5 foot-candle at feeders, drinkers, perches, and litter during daytime lighting period	Firm minimum ("must")	Lux figure below is a conversion	UEP-2024
Layer-barn light minimum in lux	Approx. 5.4 lux	Derived conversion for usability	Converted value, not printed by UEP in the layer clause	UEP-2024
Pullet-barn light minimum	0.5 foot-candles (5 lux) in all barn areas except dark-brooding spaces	Firm minimum for pullet barns	No range	UEP-2024
Perch space	6 in usable linear perch/hen	Firm minimum ("must")	No range	UEP-2024
Elevated perch share	20% of perch space elevated ≥ 16 in above adjacent floor and ≥ 1 horizontal ft from adjacent perches/walls	Firm minimum	No range	UEP-2024
Community nest space	9 sq ft/100 hens	Firm minimum	No range	UEP-2024
Individual nest ratio	1 nest/5 hens	Firm minimum	No range	UEP-2024
Feed trough space	1.5 linear in/hen with straight troughs	Firm minimum	No range	UEP-2024

Feed trough space, one-sided access	3 linear in/hen	Firm minimum	Conditional by equipment	UEP-2024
Round-pan feeder space	1.5 perimeter in/hen	Firm minimum	Conditional by equipment	UEP-2024
Bell drinkers	1 bell drinker/100 hens	Firm minimum	No range	UEP-2024
Bell-drinker circumference equivalent	0.4 circular in/hen	Firm minimum	Equipment-specific equivalent	UEP-2024
Water trough space	0.5 linear in/hen	Firm minimum	No range	UEP-2024
Water trough space, one-sided access	1.0 linear in/hen	Firm minimum	Conditional by equipment	UEP-2024
Nipple/cup drinkers	1 nipple or cup/10 hens	Firm minimum	No range	UEP-2024
Feeder travel distance	<=26 ft to feeders	Advisory/management wording ("should not need to travel more than")	No range; not clearly an auto-fail metric	UEP-2017
Drinker travel distance	<=26 ft to drinkers	Advisory/management wording ("should not need to travel more than")	No range; not clearly an auto-fail metric	UEP-2024
Backfilling/additional placement window	Hens may be added only within 20 days after initial placement; after that, the flock is established and additional birds cannot be added	Firm current UEP backfilling rule	No range	UEP-2024

Compliance interpretation: The cleanest hard compliance anchors are the system-specific floor-space minima, the audit pass threshold, and the auto-fail items. These are the most suitable variables for auto-fail tripwires. The current public cage-free guideline also defines a 20-day window for placement before a flock becomes established.

Caution: Ammonia is phrased as "should be less than 10 ppm" and "must rarely exceed 25 ppm," so it is not a simple binary continuous-monitoring rule. Feeder and drinker travel distances use "should not need to travel more than 26 feet," which reads more like a design/management expectation than an automatic audit-fail variable in the public text.

Checklist limitation: Public primary documents confirm 90% as the current passing requirement and 180/200 in the public 2017 cage-free guideline. A current publicly posted score sheet enumerating the full 200-point section-by-section structure was not located.

FDA Egg Safety Rule

Item	Firm value from primary standard	Binding status	Range / contested flag	Source
Pullet environmental test	14-16 weeks of age	Binding federal requirement	Range specified by regulation	FDA-SECG; FDA-GFI
Layer-house environmental test	40-45 weeks of age	Binding federal requirement	Range specified by regulation	FDA-SECG; FDA-GFI
Post-molt environmental test	4-6 weeks after the end of each molting process	Binding federal requirement	Range specified by regulation	FDA-SECG; FDA-GFI
Time to have egg-test results after positive environmental test	10 calendar days from notification	Binding federal requirement	No range	FDA-SECG; FDA-GFI
Egg-testing start after positive pullet environmental test	Within 2 weeks of the start of egg laying, unless eggs are diverted for life of flock	Binding federal requirement	No range	FDA-SECG; FDA-GFI
Egg-test sample size	1,000 intact eggs per test	Binding federal requirement	No range	FDA-SECG; FDA-GFI
Initial follow-up egg tests	4 tests at 2-week intervals	Binding federal requirement	No range	FDA-SECG; FDA-GFI
Total eggs represented in first egg-test series	4,000 eggs total	Binding federal requirement	No range	FDA-SECG
Monthly monitoring after flock returns to table market after earlier egg-test positivity	1 egg test/month for life of flock	Binding federal requirement	No range	FDA-SECG; FDA-GFI
Diversion/treatment requirement	Eggs must be diverted to a treatment achieving at least a 5-log destruction of SE unless the flock clears the required negative series	Binding federal requirement	No range	FDA-SECG; FDA-GFI

What counts as a positive: the retrieved FDA materials use a detection trigger rather than a numeric CFU threshold. A flock is treated as positive when the required environmental or egg test is positive for Salmonella Enteritidis using the required or equivalent method under 21 CFR 118.8. Any one positive in the required egg-test series or monthly monitoring triggers diversion obligations.

Compliance interpretation: Hard red flags include missing the 14-16 week, 40-45 week, or 4-6 week post-molt environmental testing windows; failing to begin egg testing or life-of-flock diversion after an SE-positive environmental result; failing to have egg-test results within 10 calendar days of the positive environmental notice; and failing to divert eggs after any positive egg test. The treatment standard is at least 5-log destruction of SE.

AVMA depopulation guidance

Item	Firm value or classification from primary material	Status	Range / contested flag	Source
Tier 2 definition	Tier 2 methods may be considered when emergency circumstances constrain the ability to reasonably implement a Tier 1 method or when Tier 1 methods are unavailable	AVMA guidance framework	Not numeric; framing language	AVMA-2026
Tier 3 definition	Tier 3 methods should be considered only when circumstances preclude reasonable implementation of Tier 1 or Tier 2 methods and risk from not depopulating is very high	AVMA guidance framework	Not numeric; framing language	AVMA-2026
Poultry chapter Tier 1 example	Commercial or private humane slaughter processing under 6.4.1 Tier 1 methods	AVMA guidance	No range	AVMA-Poultry
Poultry chapter Tier 2 header	6.4.2 Tier 2 methods includes inhaled methods	AVMA guidance	No range	AVMA-Poultry
Containerized gassing	Listed at 6.4.2.1.1 under Tier 2	AVMA guidance	No range	AVMA-Poultry
Whole-house gassing with gas mixtures	Listed at 6.4.2.1.2 under Tier 2	AVMA guidance	No range	AVMA-Poultry
CO2 whole-house effect example	Chickens rendered unconscious within 2 minutes of exposure to 18-20% CO2	AVMA guidance / evidence statement	Range is explicit	AVMA-Poultry
Nitrogen / gas-mixture oxygen markers	In turkeys, sensibility begins to be lost at about 9% O2, with all birds losing sensibility at 6% O2	AVMA guidance / evidence statement	Physiological marker, not a prescribed setpoint	AVMA-Poultry
VSD+ subsection	6.4.2.4.1 Ventilation shutdown plus heat and humidity	AVMA guidance; indexed as Tier 2 subsection	Classification is actively debated in the profession	AVMA-Poultry; AVMA-2026
VSD+ personnel requirement	VSD+ should only be conducted by trained and experienced personnel	AVMA guidance	No range	AVMA-Poultry
VSD+ temperature example in indexed AVMA text	House temperature rises to 104-107 F or slightly higher	AVMA indexed example / operational description	Retrieved as indexed snippet, not full directly viewable PDF passage	AVMA-Poultry

Compliance interpretation: The retrieved AVMA primary material supports the Tier 1 / Tier 2 / Tier 3 framework, with slaughter-processing pathways in Tier 1 and whole-house gassing / VSD+ heat and humidity appearing in the poultry chapter Tier 2 structure. Because direct AVMA PDF text retrieval was partial, water-based foam tier placement and any fixed humidity target should be treated as unresolved rather than inferred.

USDA APHIS HPAI response and indemnity

Item	Firm value from primary APHIS material	Status	Range / contested flag	Source
Stamping-out goal	Depopulate poultry within 24-48 hours of	Federal response goal	Goal window, not absolute guarantee	APHIS-VSD

	presumptive positive classification			
APHIS VSD+ temperature recommendation when used without CO2	104-110 F minimum, achieved as quickly as possible and preferably within 30 minutes, for at least 3 hours	APHIS operational recommendation under constrained circumstances	Not the AVMA text itself; APHIS operationalization	APHIS-VSD
APHIS constrained-circumstances approval criteria	6 criteria, including other methods not available in time and required approvals by incident management, State officials, and national incident coordinator	Federal response policy	No range	APHIS-VSD
2026 indemnity-table status	2025 VS Indemnity Table remains in effect while 2026 table is under review	Current APHIS payment status	Time-sensitive	APHIS-Indemnity
Cage-free layer indemnity, chick	\$1.12/head	Current APHIS table in effect	No range	APHIS-Table
Cage-free layer indemnity, pullet 2-17 weeks	\$5.61/head	Current APHIS table in effect	No range	APHIS-Table
Cage-free layer indemnity, Layer 1st lay 18-45 weeks	\$18.68/head	Current APHIS table in effect	No range	APHIS-Table
Cage-free layer indemnity, Layer 2nd lay 46-65 weeks	\$9.34/head	Current APHIS table in effect	No range	APHIS-Table
Cage-free layer indemnity, pre-spent hen 66-85 weeks	\$4.67/head	Current APHIS table in effect	No range	APHIS-Table
Cage-free layer indemnity, molted hen 86-115 weeks	\$4.67/head	Current APHIS table in effect	No range	APHIS-Table
Spent one-cycle hen 86+ weeks	\$0.01/head	Current APHIS table in effect	No range	APHIS-Table
Spent molted hen 116+ weeks	\$0.01/head	Current APHIS table in effect	No range	APHIS-Table

Compliance interpretation: APHIS is clearer operationally than AVMA for VSD+. APHIS ties VSD+ use to the 24-48 hour depopulation goal and, when VSD+ is used without CO2, recommends at least 104-110 F, reached preferably within 30 minutes and maintained for at least 3 hours. These are APHIS operating numbers, not animal-welfare certification metrics.

Auto-fail tripwires to encode in an evaluation

- UEP: any flock below the applicable minimum space allotment: 144 sq in/hen in multi-tier or slatted-floor housing, or 216 sq in/hen in single-level all-litter housing.
- UEP: adding birds after the 20-day placement window, evidence of backfilling, commingling certified and non-certified eggs, packing cage eggs into cage-free or organic packages, using feed-withdrawal molt, or willful abuse/neglect.
- FDA: missing a required SE environmental-test window; any SE-positive environmental test not followed by egg testing or life-of-flock diversion; any positive egg test not followed by diversion to a treatment achieving at least 5-log destruction of SE; or failure to maintain required post-clearance monthly egg testing.
- APHIS HPAI: if VSD+ is used in constrained circumstances, absence of the required incident/State/national approvals should be treated as a policy nonconformance.

Warning flags rather than automatic failure

- UEP ammonia values should be flagged as warning variables unless a conservative evaluator intentionally makes them automatic fails, because the public text uses “should” and “must rarely exceed” language.
- UEP feeder/drinker travel distances should generally be warning flags rather than auto-fails because the public text uses “should not need to travel” wording.
- AVMA VSD+ remains sensitive and actively contested in welfare discussions; the AVMA document is guidance, not a federal legal prohibition.

Open questions and limitations

The public primary materials were strong for UEP, FDA, and APHIS. AVMA retrieval was weaker because the 2026 PDF was partially blocked to direct opening. Indexed AVMA snippets established the tier framework, the placement of whole-house gassing and VSD+ in the poultry chapter, the 18-20% CO₂ / 2-minute unconsciousness example, and the 104-107 F VSD+ temperature example. A clean directly viewable AVMA passage fully enumerating the water-based-foam subsection header or fixed humidity target was not located.

The UEP 200-point audit checklist structure is also incomplete from public sources. Public primary materials confirm 90% as the current pass requirement and 180/200 in the 2017 cage-free guideline, but a current publicly posted UEP score sheet breaking the checklist into scorable sub-sections was not located.

Source key

UEP-2024: UEP Certified, 2024 Cage-Free Housing Guidelines: https://uepcertified.com/wp-content/uploads/2023/10/CF-UEP-Guidelines_2024.pdf

UEP-2017: UEP Certified, 2017 Cage-Free Animal Husbandry Guidelines: https://uepcertified.com/wp-content/uploads/2021/08/CF-UEP-Guidelines_17-3.pdf

FDA-SECG: FDA, Small Entity Compliance Guide: Prevention of Salmonella Enteritidis in Shell Eggs During Production, Transportation, and Storage: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/small-entity-compliance-guide-prevention-salmonella-enteritidis-shell-eggs-during-production>

FDA-GFI: FDA, Guidance for Industry: Prevention of Salmonella Enteritidis in Shell Eggs During Production, Storage, and Transportation: <https://www.fda.gov/files/food/published/Guidance-for-Industry--Prevention-of-Salmonella-Enteritidis-in-Shell-Eggs-During-Production--Storage--and-Transportation-PDF.pdf>

FDA-Method: FDA, Environmental Sampling and Detection of Salmonella in Poultry Houses: <https://www.fda.gov/food/laboratory-methods-food/environmental-sampling-and-detection-salmonella-poultry-houses>

AVMA-2026: AVMA, Guidelines for the Depopulation of Animals: 2026 Edition: <https://www.avma.org/resources-tools/avma-policies/avma-guidelines-depopulation-animals>

AVMA-Poultry: AVMA, 2026 Poultry chapter PDF/indexed primary text: <https://www.avma.org/sites/default/files/2026-01/CHAPTER6-POULTRY-INTRO.pdf>

APHIS-VSD: USDA APHIS, HPAI Response: Ventilation Shutdown Plus Policy: <https://www.aphis.usda.gov/sites/default/files/ventilationshutdownpolicy.pdf>

APHIS-Indemnity: USDA APHIS, Producer Indemnity and Compensation: <https://www.aphis.usda.gov/livestock-poultry-disease/epidemiology/producer-indemnity-comp>

APHIS-Table: USDA APHIS, Veterinary Services Indemnity Table for 2025: <https://www.aphis.usda.gov/sites/default/files/vs-indemnity-table.pdf>